Mihir Sharma, Independent researcher. IUI 2027 Posters

Can a Buzz Answer That?

How much of what blind people ask a visual assistant fits a vibration vocabulary

Four quick pulses mean “I can’t tell”: the buzz a vibration channel should send most often.

15.5% of 8,000 held-out questions from blind people could get an informative buzz



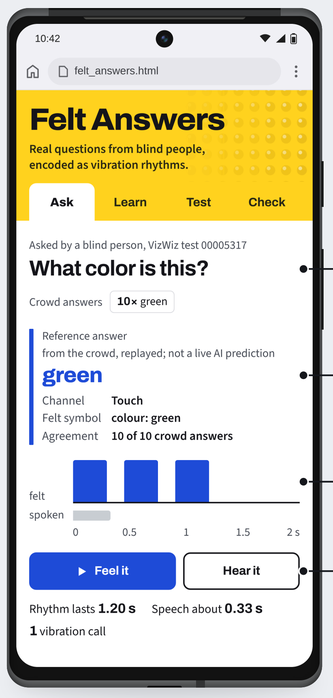
- **10.9%** buzz, at least half agree
- **4.6%** buzz, then the unsure flutter
- **26.8%** “I can’t tell”
- **57.7%** speech: no symbol fits

42.3% could be felt in all. Answers: colour 8.6, no 2.6, yes 2.4, number 1.9%.

Encoded, not yet communicated: no blind or low-vision person has felt these rhythms yet.

The prototype: Felt Answers

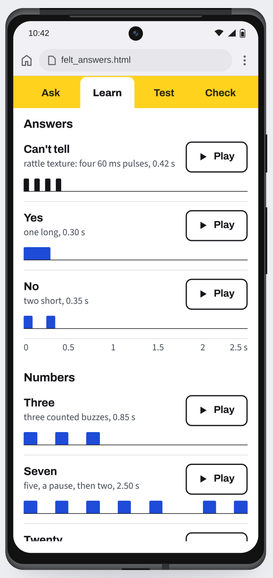
One offline web page for Chrome on Android, built as a **research instrument** with four modes. Real screens, replaying crowd answers; no live AI. It matches the analysis on all 32,842 questions; rendering on phone motors is not yet verified.



One complete interaction, as a visitor will try it at the poster:

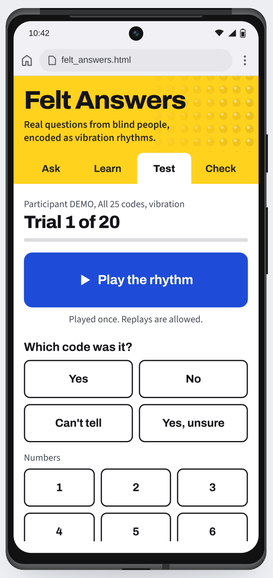
1. **A real VizWiz question**, with its ten crowd answers
2. **Reference answer**, replayed from the crowd: green, 10 of 10 agree
3. **Three long pulses** mean green: 1.2 s, one vibration call
4. **Feel it** or hear it; speech would take about 0.33 s

Ask



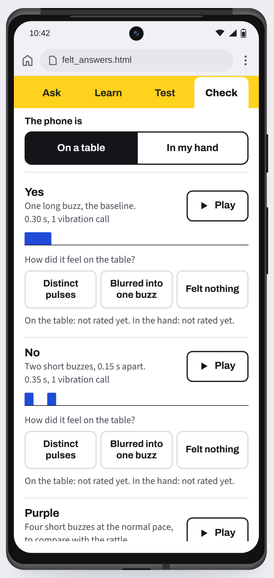
Learn

Every code with its rhythm and a Play button: can’t tell, yes, no, numbers, 11 colours, unsure.



Test

Identification trials in shuffled blocks; the answer stays hidden until a response. Logs time and replays; confusion matrix, CSV export.

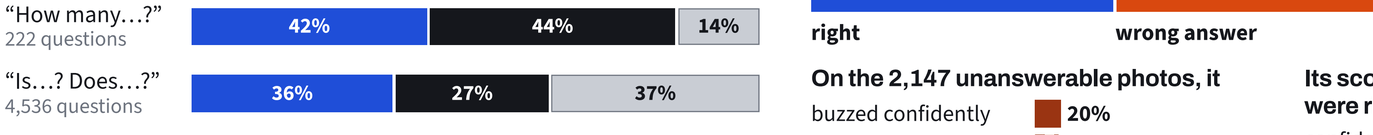


Check

Before any participant: rate each rhythm on a table and in the hand, and log when each vibration call fires.

How every reply is routed

32,842 photo questions from blind people (VizWiz[1]), 10 crowd answers each.[2–4] The crowd’s most frequent answer stands in for an assistant. Four real test questions, one per route; shares are of all 8,000.



All rules fixed before the test split was opened.

Could people answer it? (VizWiz flag) — no → **Buzz “can’t tell”** — 26.8%
“What is the expiration date on this nose spray?”
0.42 s; unanswerable ×10

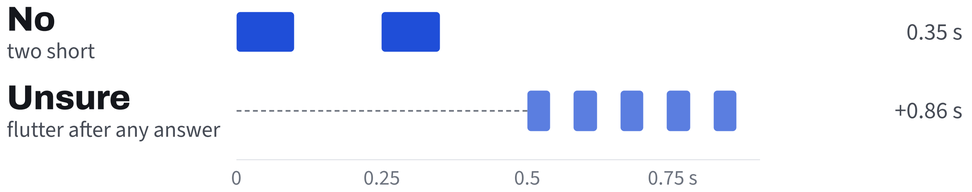
yes → **Does the answer have a symbol?** — no → **Speak it** — 57.7%
“What is in this can?”
kidney beans ×3, beans ×3, red kidney beans ×2, …: only 3 of 10 match word for word

yes → **Do at least 5 of 10 answers give that symbol?** — no → **Buzz, then unsure flutter** — 4.6%
“What color are these pants?”
white ×4, tan ×3, khaki ×2, unanswerable ×1
white, then flutter; 4 of 10

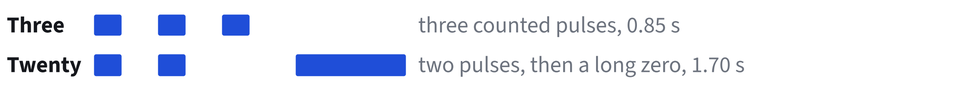
yes → **Buzz the answer** — 10.9%
“What’s the value of this bill?”
one pulse: “1 dollar” ×4, “1” ×4, “$1”: 9 of 10 give this symbol

A tactile alphabet

Rhythm is the most reliably felt tacton dimension.[5] Four core codes, to scale:



Plus numbers and eleven colours

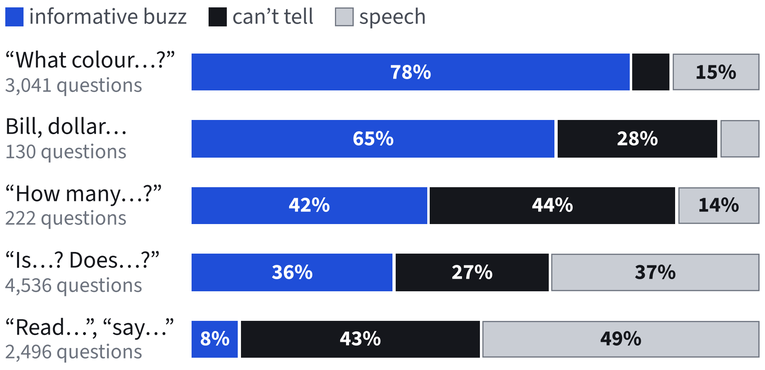


Colours are three- or four-element codes: white sss, black sls, green lll, purple ssss, and seven more. **s** 100 ms, **l** 300 ms, 150 ms gaps; the rattle and flutter use faster pulses, untested on phone motors.

Open question. No and two, yes and one differ only in timing. Wording predicted the kind of answer for 83% of felt answers, so context *might* tell them apart.

What you ask decides

Colour and money questions fit a buzz; reading rarely does.



Keyword groups across all 32,842 questions. Even 128 learned codes lift felt replies only to 55.5%.[6, 8]

A real model rarely says “can’t tell”

ViLT,[11] an older open model not trained on VizWiz or to abstain, answered all 8,000 test questions; a buzz is confident if its score is at least 0.5.

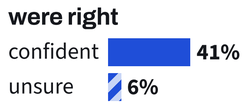
32% of its 1,867 confident buzzes were right
answerable photo, and at least 3 of 10 crowd answers give that symbol



On the 2,147 unanswerable photos, it



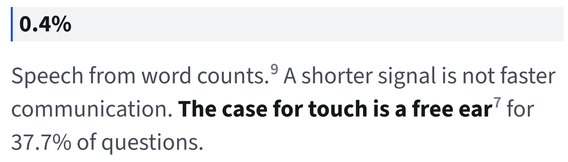
Its score carries some signal: buzzes that were right



*Unplanned count of answers such as “unknown”. Score barely separates unanswerable photos (AUROC 0.57). A probe of a failure mode, not a benchmark of today’s assistants.[10]

Touch is not reliably faster

Felt replies that a buzz delivers before speech



Speech from word counts.[9] A shorter signal is not faster communication. **The case for touch is a free ear**[7] for 37.7% of questions.

What this means for design

Touch for failure, doubt and closed answers
“Can’t tell” is the most common buzz (26.8%). 29% of felt answers need the flutter. Closed answers agree most: yes 79%, colour 66%, spoken 42% word for word.

Speech for reading, naming and describing
57.7% of questions; only 8% of “Read…” questions could be felt.

Abstention first
A can’t-tell buzz is only useful if the assistant knows when it can’t tell. **It must detect unanswerable photos and say so.**

Next: feel it

1. **Device check** — rendering and call timing (Check)
2. **Ethics approval** — no participant study run yet
3. **Identify every code** — blind and low-vision people (Test)
4. **Touch against speech** — time and preference, in noise

Encoded is not yet communicated.

Let’s discuss

1. When should an AI assistant communicate uncertainty through touch rather than speech?
2. How small should an accessible tactile vocabulary be?
3. What evidence would show that tactile answers improve the experience of blind and low-vision people?

Feel the rhythms on the phone at the poster.

Data. VizWiz-VQA and its answer-difference, image-quality and skill labels (CC BY 4.0); code, data and prototype are in the supplement. **AI use.** Claude (Anthropic) wrote and ran the code and drafted the paper and this poster.

1. Gurari et al. VizWiz Grand Challenge. CVPR 2018.
2. Bhattacharya et al. Why does a visual question have different answers? ICCV 2019.
3. Chiu et al. Assessing image quality issues for real-world problems. CVPR 2020.
4. Zeng et al. Vision skills needed to answer visual questions. CSCW 2020.
5. Brown et al. A first investigation into the effectiveness of tactons. WHC 2005.
6. Azadi & Jones. Evaluating vibrotactile dimensions for the design of tactons. ToH 2014.
7. Giudice & Legge. Blind navigation and the role of technology. Wiley 2008.
8. Chase et al. Learning vibes. WHC 2021.
9. Bragg et al. A large inclusive study of human listening rates. CHI 2018.
10. Whitehead et al. Reliable visual question answering. ECCV 2022.
11. Kim et al. ViLT: vision-and-language transformer without convolution or region supervision. ICML 2021.